MACHINE LEARNING-BASED WIRELESS INTRUSION DETECTION

Synthetic-data prototype for detecting deauthentication-flood-like traffic

WIDS | Python • scikit-learn • Random Forest • Google Colab

2. SYSTEM ARCHITECTURE

Synthetic traffic generator

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Feature extraction (5 features)

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Train/test split (75% / 25%)

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Random Forest classifier

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Metrics + confusion matrix

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Alert output

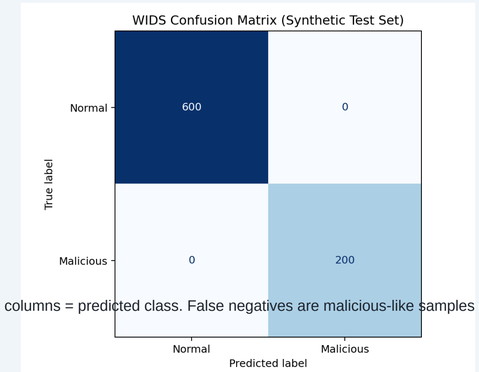
Features: frame type code, simulated RSSI, frames per second, deauthentication frames per second, and unique destination count.

3. DATA & MODEL

Dataset: 3,200 synthetic rows — 2,400 normal and 800 malicious-like. The Random Forest uses 120 trees, maximum depth 8, and a fixed random seed.

Labels: 0 = normal; 1 = malicious-like. Evaluation uses a stratified held-out test split.

4. CONFUSION MATRIX



Rows = actual class; columns = predicted class. False negatives are malicious-like samples classified as normal.

5. RESULTS

Synthetic test accuracy: 100.00%

Confusion matrix [[TN, FP], [FN, TP]]: [[600, 0], [0, 200]]

See results/metrics.json for precision, recall, F1-score, and the complete classification report.

6. LIMITATIONS & NEXT STEPS

Synthetic data may make classes unrealistically easy to separate. These scores do not demonstrate real-world detection performance.

Next: collect authorised monitor-mode captures; label benign and attack windows; test across devices; measure false positives; evaluate time-based splits; add robust thresholds and drift monitoring.

AI ASSISTANCE

AI coding assistance helped draft and organise code and documentation. The author should review and explain the feature engineering, train/test split, classifier, metrics, and alert logic.